Rochester Institute of Technology

Preserving Image Quality in Home Projection

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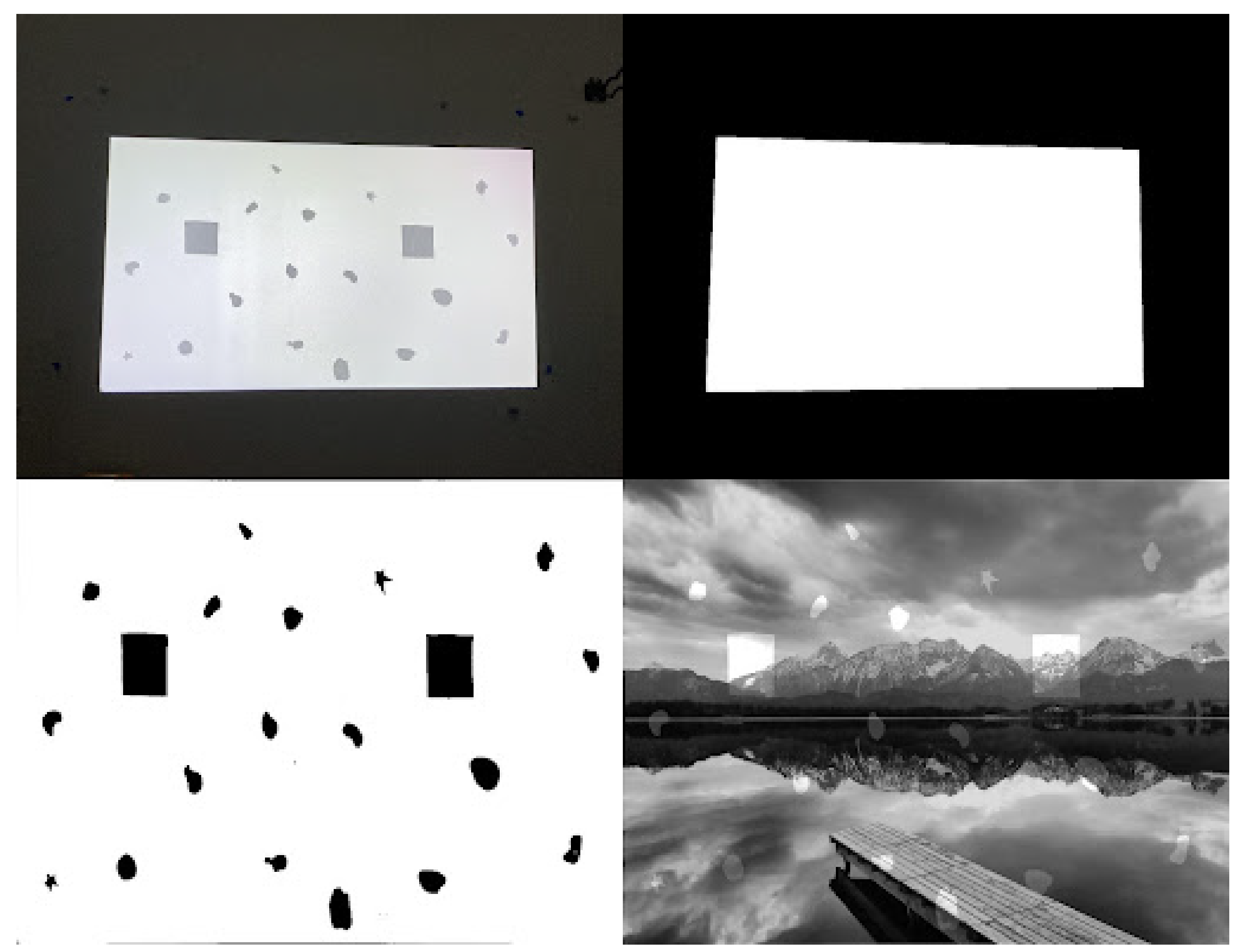
Problem:

Projecting images onto non-uniform screens, like patterned walls, will look uneven and can negatively impact the viewing experience. The goal of this project is to assess the impact of these surface imperfections and adjust the projected image so it looks as good as possible on any wall or screen.

Luminance Correction:

First, the surface is analyzed to detect any imperfections. The image is then adjusted based on the maximum and minimum reflectances of the non-uniform screen, to create a uniform image where imperfections are no longer visible. Methods of rolling off the highlights and shadows are also applied



Pipeline for projection screen correction: from original projection screen capture to screen isolation, mapping imperfections, and surface-aware image correction.

Image Quality Evaluation:

Should we clip the image to achieve perfect uniformity, or preserve some highlights and shadows at the expense of uniformity? The final phase of the project studies observer preferences to find the best balance.

Full correction and partial correction side by side